PORTFOLIO

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Computer Science · Software Developer

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Education	Computer Science
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> I design and ship desktop applications and data pipelines with concrete deliverables: working demos, screenshots, and reports clients can open and review.

Work

1. Cosmic Observer

Windows astronomy desktop app — 3D sky, NASA data, AI tutor.

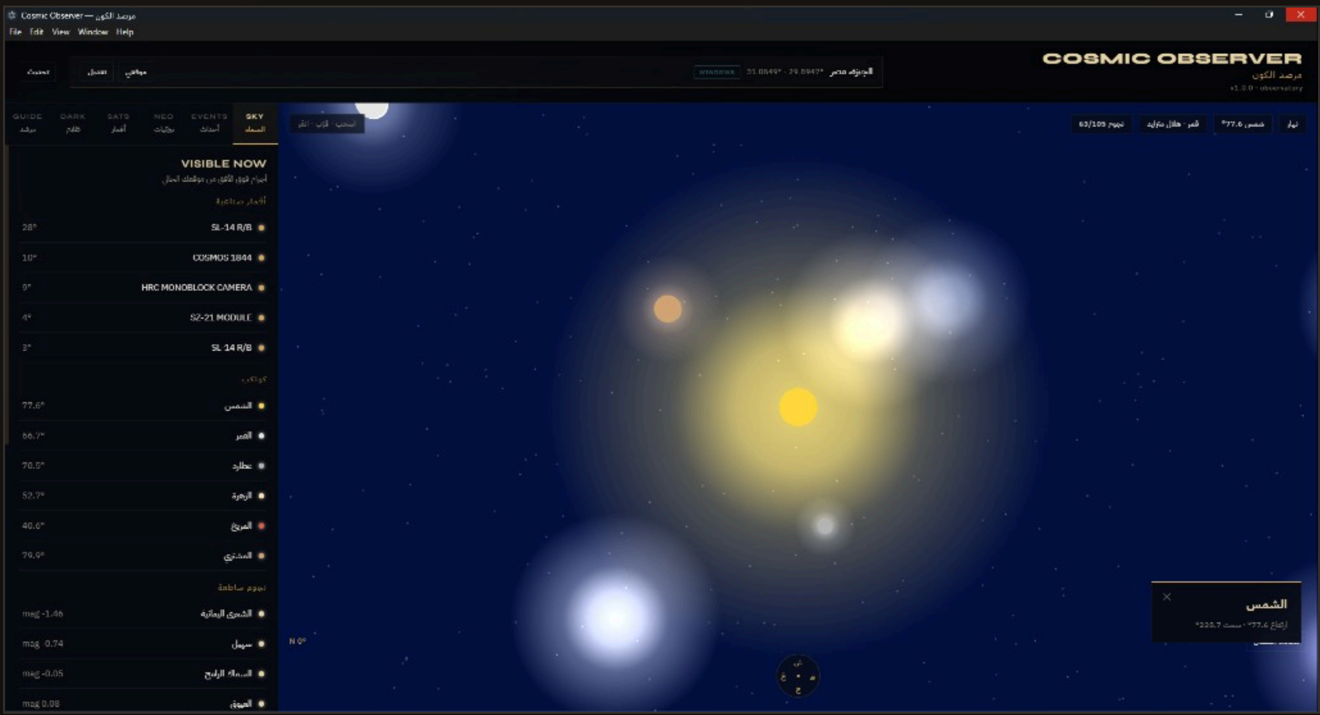
2. Vertical Farm Analysis

Python pipeline — farm telemetry to charts & PDF report.

PROJECT 01

Cosmic Observer

A Windows desktop astronomy application with an interactive 3D sky, near-Earth object tracking, orbital debris, a dark-matter lensing research module, and an AI astronomy tutor — packaged as a native .exe.



What you see

- Live 3D sky from GPS / manual location
- Visible satellites, planets, and bright stars
- Sun / Moon status and sky events
- Tabs: Sky, Events, NEO, Sats, Dark, Guide
- Bilingual interface (Arabic + English)

Repository

- GitHub: github.com/engboda6/cosmic-observer
- License: MIT
- Platform: Windows .exe
- Demo video on the portfolio site

PROJECT 01 · STACK

Cosmic Observer — Technical Detail

JavaScript end to end: React and Three.js for the interface and sky, Node.js/Express for local APIs, Electron for the Windows shell.

Tech stack

JavaScript | React 18 | Three.js | Vite | Node.js / Express | astronomy-engine | OpenAI SDK | Electron 33 | GLSL | PowerShell

Highlights

- **3D Sky** — stars, planets, Sun, Moon
- **Sky Events** — sunrise/sunset, moon phases
- **NEO Tracking** — NASA NeoWs
- **Orbital Debris** — CelesTrak TLE
- **Dark Matter** — research-style lensing maps
- **AI Tutor** — OpenAI chat (optional key)

Data sources

- NASA NeoWs
- JPL Sentry / CAD
- NASA CNEOS Fireball
- CelesTrak (TLE)
- VizieR / CDS

Architecture

- `electron/` — main + preload
- `server/` — APIs
- `src/` — React + Three.js

PROJECT 02

Vertical Farm Telemetry Analysis

A Python analysis system for a vertical farm. Reads a real telemetry ledger with 600 production rows and produces a clean dataset, analysis tables, charts, Excel exports, and a delivery-ready PDF report.

Pipeline

1. **Import & clean — prepare_dataset.py**
   Reads the telemetry CSV, cleans columns, parses Stack / Rack / Row IDs.

2. **Analysis — analysis.py**
   Nutrient decay matrix for 60 channels, weak circulation, harvest index, feeding balances.

3. **Charts — charts.py**
   EC distribution, circulation vs decay, heatmap, harvest index.

4. **Report — report.py**
   Three-page PDF by Abd Alrahman Ahmed: summary, matrix, recommendations.

5. **Full run — run_pipeline.py**
   Executes every step with one command.

PROJECT 02 · DELIVERY

Vertical Farm — Stack & Contact

Language: Python only. Raw farm CSV → structured dataset, analysis tables, charts, and PDF.

Libraries

- **pandas** — read, clean, aggregate
- **numpy** — calculations
- **matplotlib** — charts + PDF
- **openpyxl** — Excel export

Deliverables

- Clean structured dataset
- Analysis for 60 irrigation channels
- Charts and Excel files
- Three-page PDF report

Contact

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